IT Support Final Project

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Windows Server & IT Support Practical Lab

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Project Overview

This project was designed to simulate a real-world IT Support environment using Windows Server and Windows 11 virtual machines.

The project covered multiple practical IT support tasks including Active Directory deployment, user account management, Group Policy configuration, printer troubleshooting, remote administration, permissions management, firewall configuration, backup and restore operations, and DNS troubleshooting.

The objective of this project was to gain hands-on experience in diagnosing, troubleshooting, and resolving common IT infrastructure problems while following professional documentation practices.

Lab Environment

The lab environment consisted of a Windows Server machine configured as a Domain Controller and DNS Server, along with a Windows 11 client machine joined to the domain.

The environment was deployed using Oracle VirtualBox to simulate a small enterprise network for hands-on IT support training and troubleshooting.

Lab Components:

- Windows Server
- Active Directory Domain Services (AD DS)
- DNS Server
- Windows 11 Client
- VirtualBox
- Remote Desktop Services
- Shared Folders and NTFS Permissions.

Task 1 - Active Directory Deployment

Company OU Structure

To simulate a real company environment, a dedicated Organizational Unit (OU) named "Fekri Company" was created within Active Directory.

Multiple departmental OUs were then created to represent the company's structure, including Management, Human Resources, Finance, IT Support, Sales, and Marketing.

This structure allows efficient administration of users, groups, policies, and resources based on departmental requirements.

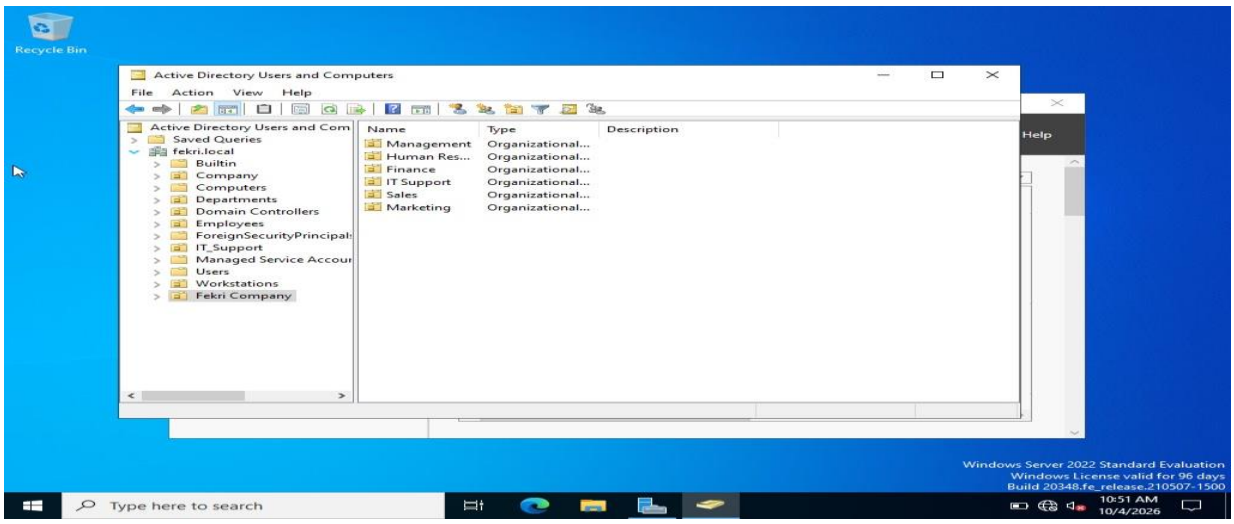


Figure 1 - Organizational Unit Structure for Fekri Company

Department User Accounts

User accounts were created for each department within the company structure.
Each department contains dedicated users to simulate a real enterprise environment and to support permission management, security groups, and Group Policy deployment
The following example shows the user accounts created for the IT Support department.

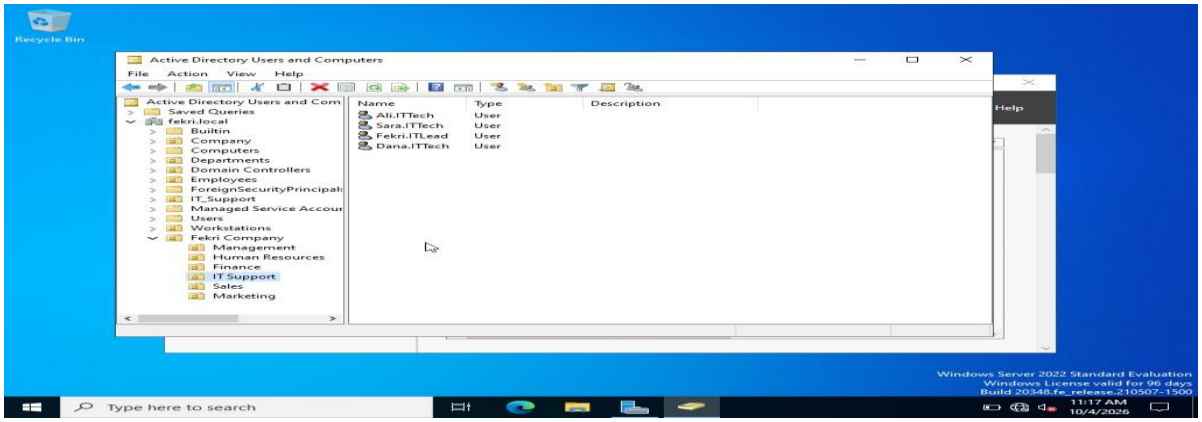


Figure 2 - IT Support Department User Accounts

Security Groups

Security groups were created to simplify permission management and resource access within the organization.

Instead of assigning permissions directly to individual users, users were added to departmental security groups. This approach follows Active Directory best practices and simplifies administration.

The IT_Support_Staff security group was created to manage access rights and permissions for the IT Support department.

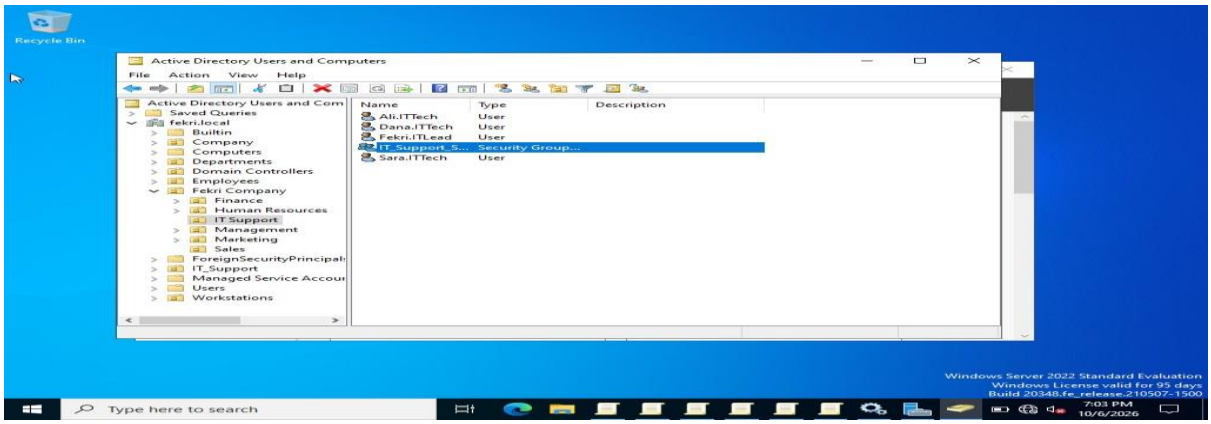


Figure 3 - IT Support Security Group

Shared Folders and Permissions

Departmental shared folders were created to provide centralized file storage for each department within the company.
Separate folders were created for Finance, Human Resources, IT Support, Management, Marketing, and Sales to simulate a real-world enterprise file-sharing structure.
This approach improves organization, security, and access management across departments.

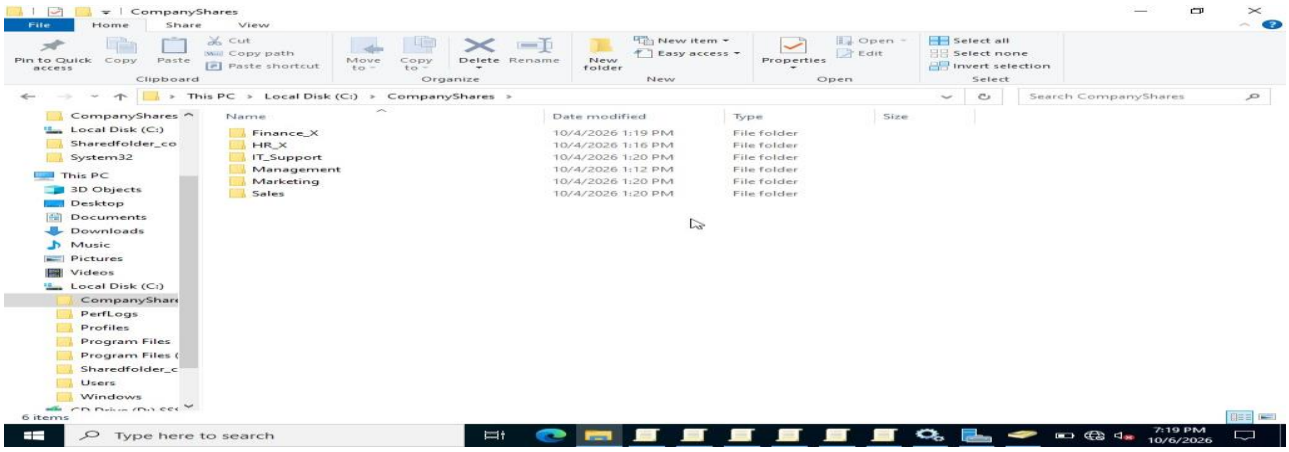


Figure 4 - Department Shared Folders Structure

Shared Folder Permissions

Folder sharing permissions were configured using departmental security groups.
The IT_Support_Staff security group was assigned the appropriate access rights to the IT Support shared folder, ensuring that permissions could be managed through Active Directory groups rather than individual user accounts.
This follows security best practices and simplifies administration.

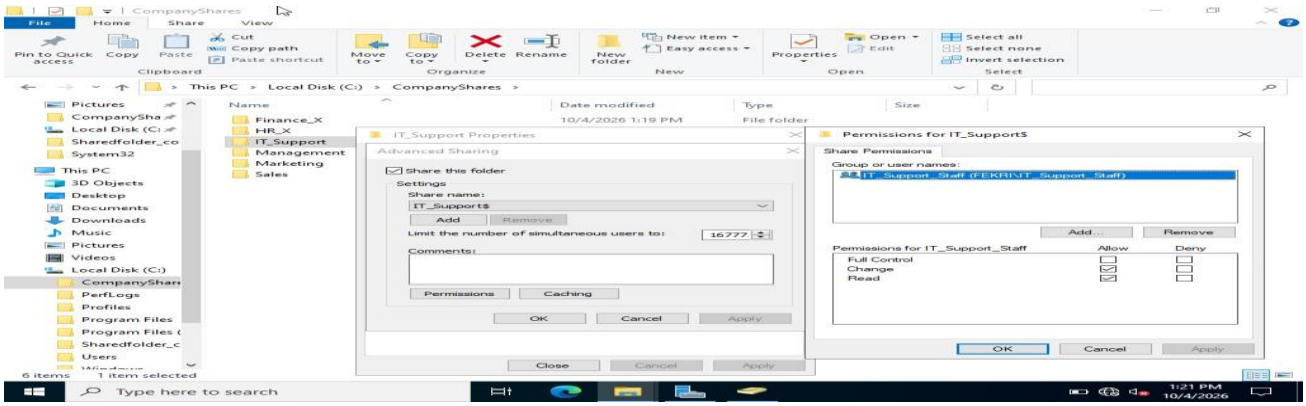


Figure 5 - IT Support Shared Folder Permissions

Printer Deployment and Management

A shared network printer was configured and deployed within the Active Directory environment.
The printer was shared on the server and assigned appropriate security permissions to control access for authorized users.
To simplify printer deployment and management, Group Policy Preferences (GPP) were used to automatically deploy the shared printer to client computers within the domain.
This approach ensures consistent printer availability and reduces manual administrative effort.

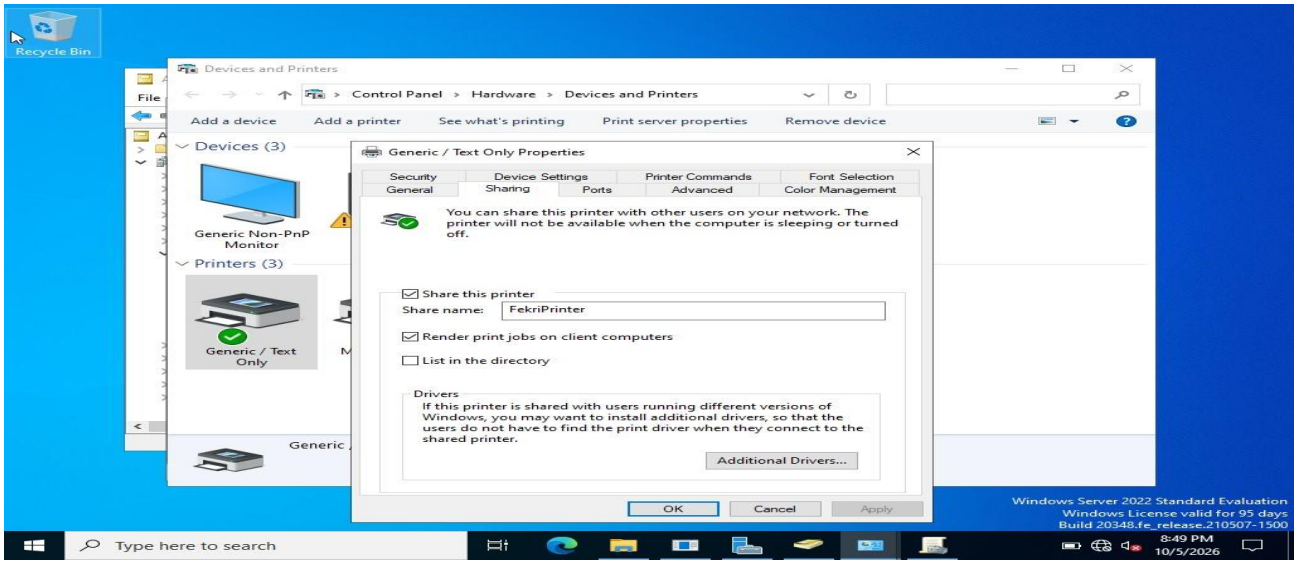


Figure 6 - Shared Printer Configuration

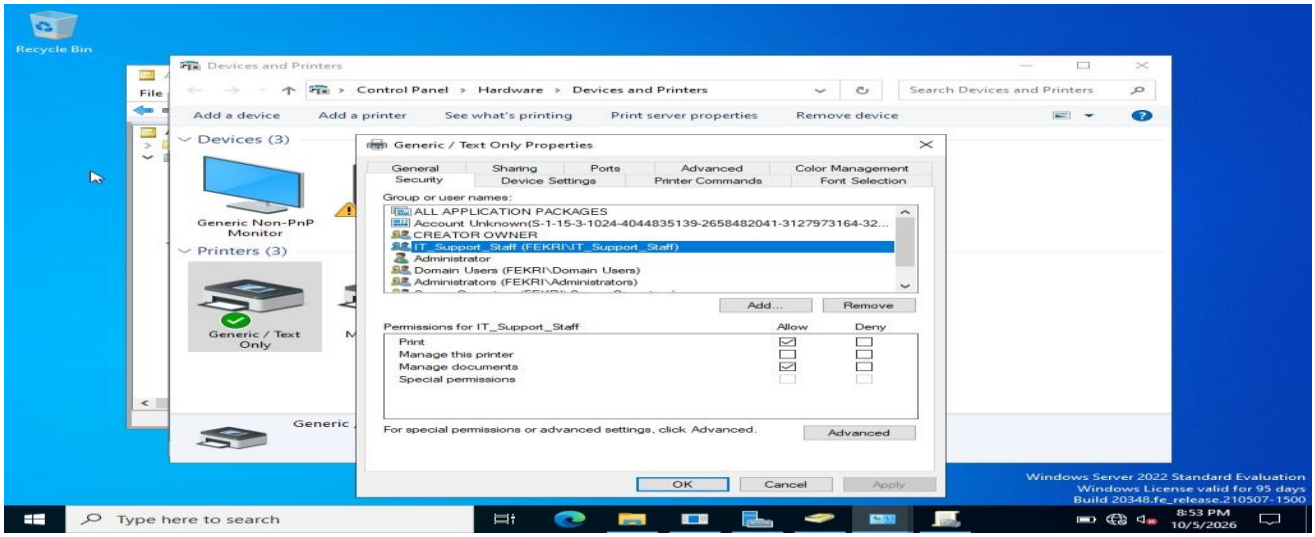


Figure 7 - Printer Security Permissions

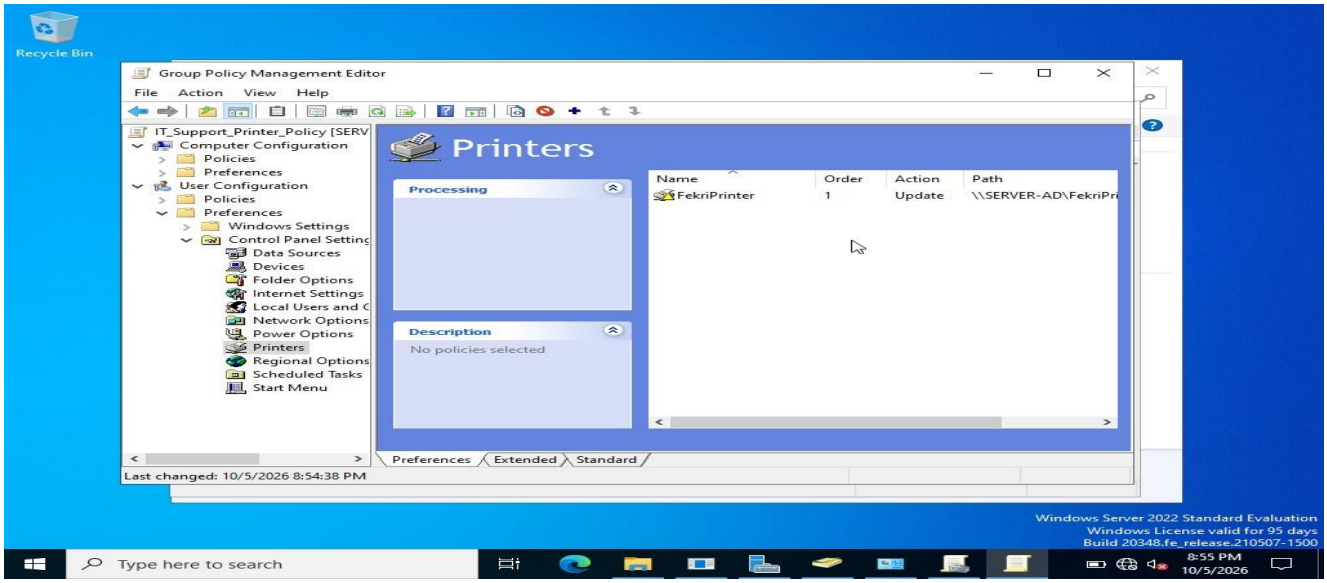


Figure 8 - Printer Deployment Using Group Policy.

Client Validation and Access Testing

Client-side testing was performed to verify connectivity, resource availability, and access permissions within the Active Directory environment.
Users were able to access only the resources assigned to their department through security group membership and configured permissions.
Hidden administrative shares and departmental folders were successfully accessed from the client workstation, confirming correct configuration of network sharing and access controls.
This testing phase verified that the implemented security model was functioning as intended.

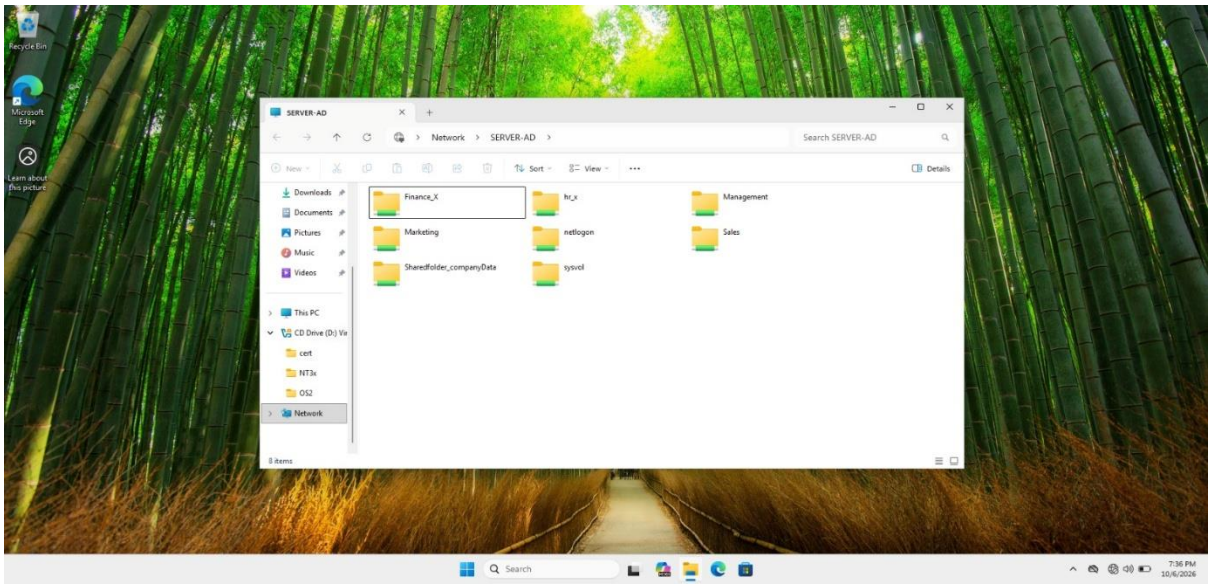


Figure 9 - Client Access Verification to Shared Resources.

Tickets and Troubleshooting

Ticket 1 - Finance Share Access Issue

Issue
A user was unable to access the Finance department shared folder from the client workstation.
Investigation
Access permissions and group membership were reviewed to determine the cause of the issue.
Root Cause
The user account was not a member of the appropriate security group required to access the Finance department shared folder.
Resolution
The user was added to the correct Active Directory security group and permissions were validated.
Result
The user was successfully able to access the Finance shared folder after group membership was updated.

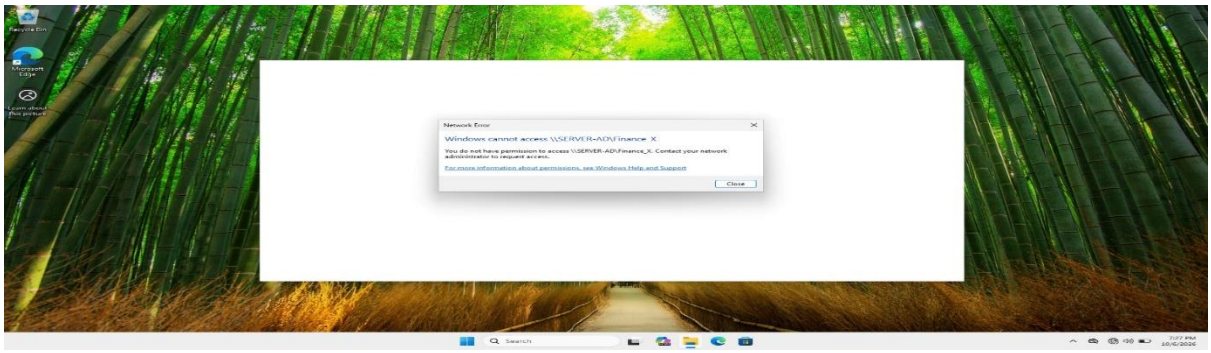


Figure 10 - Access Denied Error Before Resolution

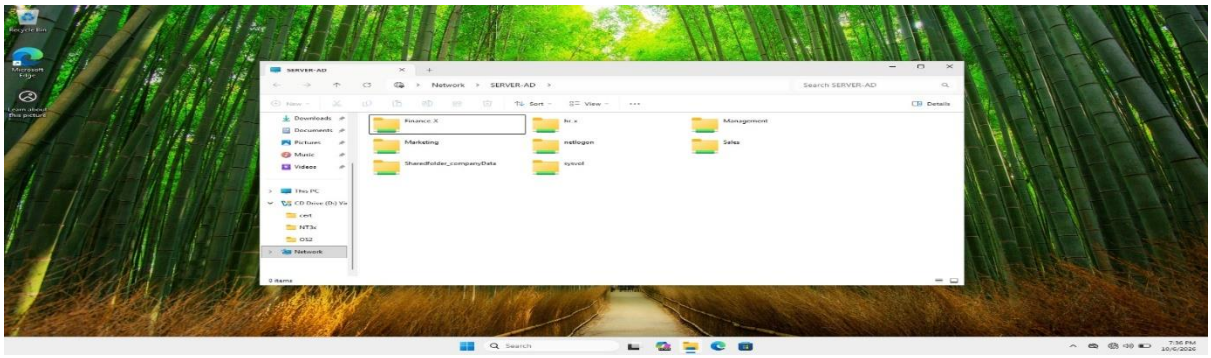


Figure 11 - Successful Access After Group Membership Update

Ticket 2 - Network Printer Access Issue

Issue
A domain user was unable to access and use the shared network printer from the client workstation.
Investigation
Network printer configuration, printer sharing settings, and deployment policies were reviewed to identify the source of the problem.
Root Cause
The shared printer was not being properly accessed from the client computer, resulting in connection and availability issues.
Resolution
Printer sharing settings, deployment configuration, and client connectivity were verified and corrected. The printer deployment process was then retested.
Result
The shared network printer became accessible from the client workstation and printing functionality was successfully restored.

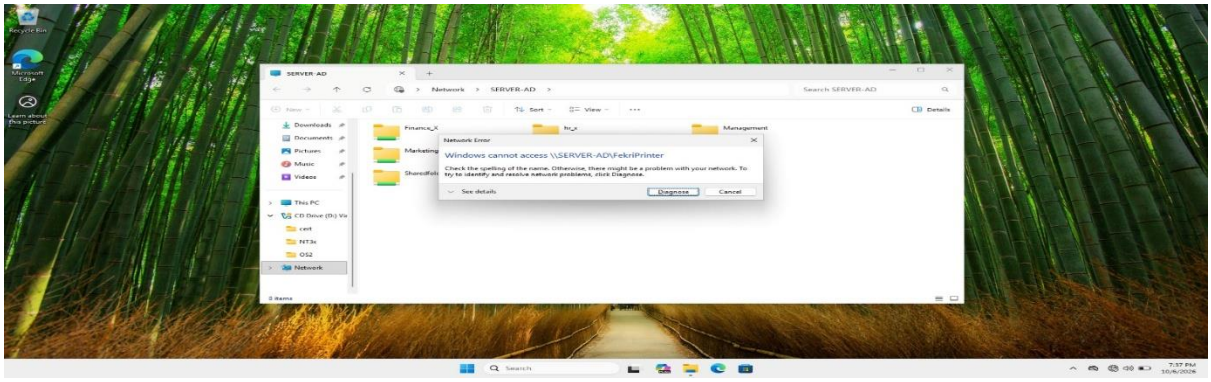


Figure 12 - Network Printer Access Error Before Resolution

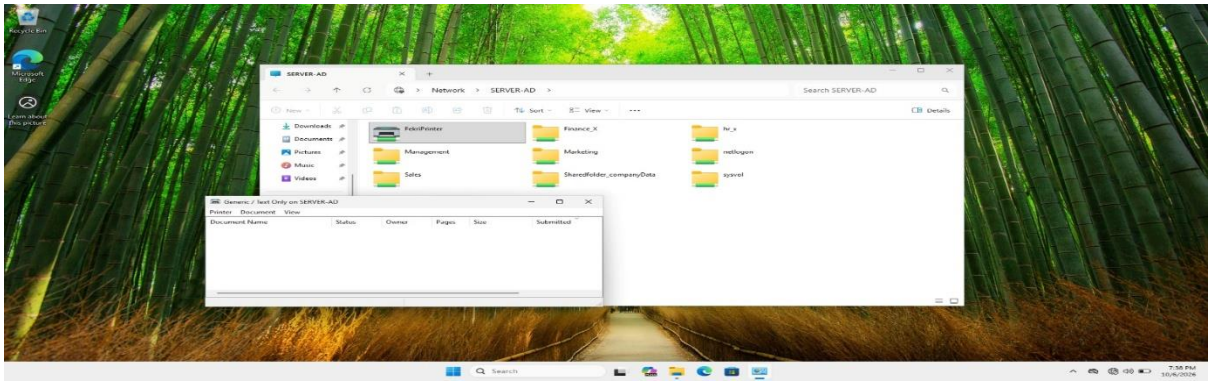


Figure 13 - Successful Printer Access After Resolution

Ticket 3 - DNS Resolution Issue

Issue
A domain client was unable to properly resolve the domain controller hostname using DNS.
Investigation
Network connectivity was tested using Ping and DNS name resolution was tested using Nslookup.
Root Cause
The client was configured to use an incorrect DNS server, preventing proper hostname resolution within the Active Directory environment.
Resolution
The DNS configuration was corrected to point to the domain DNS server and name resolution was revalidated.
Result
The client was able to successfully resolve the domain controller hostname and Active Directory name resolution functionality was restored.

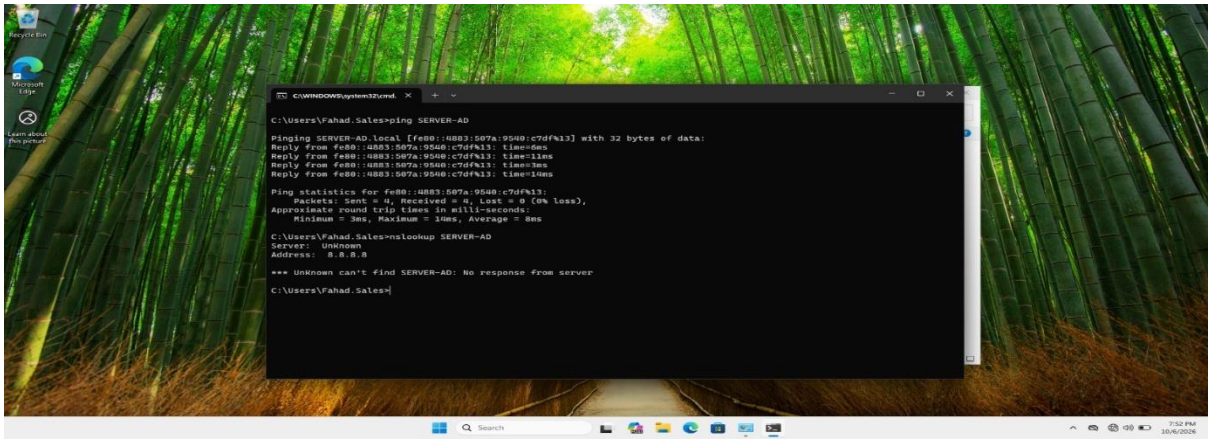


Figure 14 - DNS Resolution Failure Before Correction

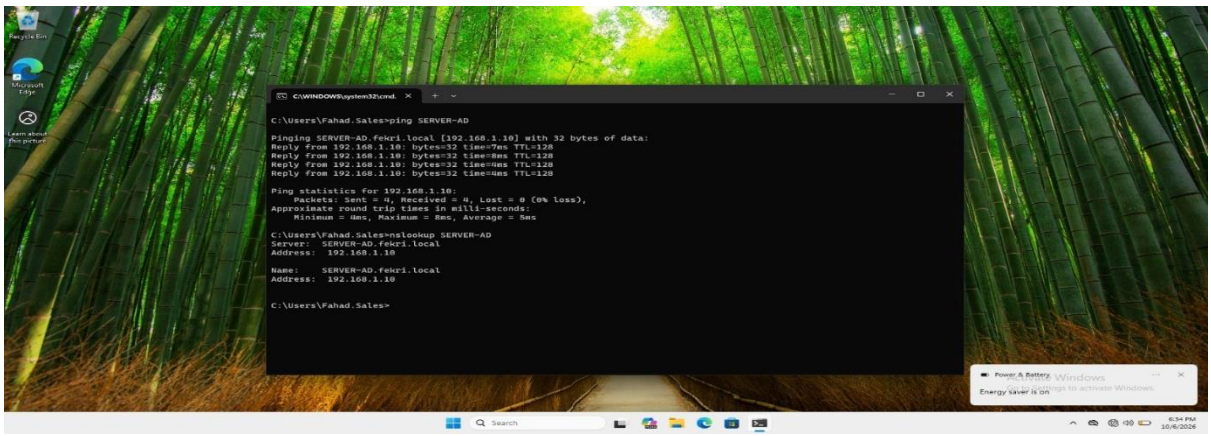


Figure 15 - Successful DNS Resolution After Correction

Final Project Conclusion

This project successfully demonstrated the deployment and administration of a complete Active Directory environment within a virtual enterprise infrastructure.

The implementation included Organizational Units (OUs), user and group management, shared folders, NTFS and share permissions, network printer deployment, and Group Policy configuration.

Client-side validation confirmed successful access control, printer deployment, resource availability, and domain functionality.

Additionally, multiple real-world troubleshooting scenarios were performed, including shared folder access issues, network printer connectivity problems, and DNS resolution failures. Each issue was investigated, resolved, and validated using industry-standard troubleshooting methodologies.

This project provided practical hands-on experience in Active Directory administration, IT Support operations, user access management, network services, and problem resolution.

The completed environment reflects a realistic small business infrastructure and serves as a professional portfolio project demonstrating core IT Support and Systems Administration skills.